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A Unique Priority Aware Adaptive Optimization Protocol for Enhancing Lifetime in Heterogeneous IoT Based Wireless Sensor Networks

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ABSTRACT

Energy efficiency is a critical factor in enhancing the operational lifetime of Wireless Sensor Networks (WSNs), particularly in Internet of Things (IoT) applications where limited battery resources constrain sensor nodes. This paper proposes an enhanced protocol, Energy-Efficient Power-Efficient Gathering in Sensor Information Systems with Power-Aware Hierarchical Clustering (EEPEG-PA-H), and presents a comparative evaluation against the Simple Fractional Routing Protocol (SFRT). The EEPEG-PA-H protocol integrates adaptive dynamic clustering with power-aware hierarchical routing, enabling real-time adjustments in cluster formation and routing paths based on residual energy, traffic load, and network conditions. In contrast to the static nature of SFRT, the dynamic strategy of EEPEG-PA-H significantly reduces redundant transmissions and balances energy usage across nodes. Simulation results indicate that EEPEG-PA-H improves network lifetime by 23.33% over PEGASIS and by 1.34% over the earlier EEPEG-PA model, while maintaining a PDR of 76% and a throughput of 1.2 Mbps. This improvement is achieved by significantly reducing the End-to-End Delay to 190 ms and the Packet Loss Rate to 22%. Furthermore, it demonstrates superior performance compared to SFRT, LEACH, and VSSLS-SIACR in terms of energy consumption and network sustainability. The results demonstrate that EEPEG-PA-H provides a robust and energy-efficient routing framework for heterogeneous WSNs in IoT applications, with potential applicability in dynamic environments such as mobile healthcare.

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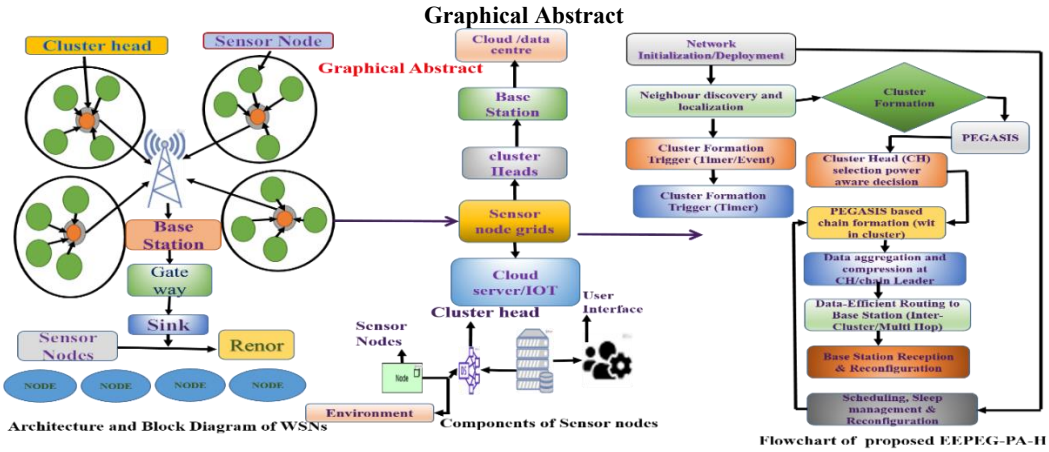
Graphical Abstract



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1. INTRODUCTION

A Wireless Sensor Network (WSN) is a communication network involving several wireless nodes and interlinks that allows the users to connect in a reliable, easy, and expandable manner. A single machine in the network performs both the functions of a router and a client. All transfers of information from any machine to another will be on an as ordered basis, and all this will be dynamic and self-organizing. This type of topology improves the fault tolerance capacity of the network, since in case any node is down or not within reach, other nodes can be utilized to reroute the data to the intended end user. This type of network is also known to be used in cities, rural areas where there is no conventional infrastructure, or during disaster recovery operations. These networks also allow for some Wireless standards; therefore, users do not need a router to connect devices like wifi. Therefore, the use of WSNs may be explored as a strategy to enhance network coverage and facilitate improvement in communication efficiency (1).

The integration of VoIP with a WSN creates an effective medium for voice transmission using a non-centralized wireless architecture. Wireless Sensor Networks (WSNs) consist of a large number of distributed sensor nodes that monitor and gather data on environmental and physical conditions, including temperature, humidity, motion, and pressure. These nodes communicate wirelessly, transmitting the collected data to a central base station for analysis and informed decision-making (2). The WSN lies in its ability to provide real-time, remote monitoring and automation across a wide range of applications. WSNs enable intelligent and autonomous systems by offering scalability, flexibility, and low-cost distribution, especially in inaccessible environments (3). With the growing integration of the Internet of Things, WSNs have become a fundamental building block, facilitating consistent communication between devices and enabling data-driven insights (4). Energy efficiency and reliability in WSNs are critical, as sensor nodes typically operate on limited battery power, and maintaining long-term, uninterrupted performance is essential for their effectiveness (5). With their versatility and wide range of applications, WSNs face several limitations that can affect the performance and deployment of network nodes (6). Sensor nodes are battery-powered, and frequent battery replacement is impractical in many scenarios due to energy constraints that limit processing power, communication range, and sensing capabilities (7). WSNs must be scalable to effectively manage large-scale deployments consisting of hundreds or even thousands of nodes, as managing such complex networks can be challenging (8).

In any WSN design, each node operates both as a client and a router. This gives the capability of switching voice data packets (VDP) through various alternative routes around the network. In such a scenario, even when a single node fails or is interfered with, there is no interruption in communication. The VoIP over wireless mesh network application is particularly useful in regions where there are no wires for communication, because it is suitable for remote places or for emergencies (9). The use of self-healing mesh technologies helps to enhance call quality by adaptively routing calls to achieve minimal delay, jitter, and packet loss, which are crucial for better voice quality. An increased network size can lead to higher data congestion, latency, and reduced coordination efficiency. The network nodes have a short communication range, often requiring multi-hop routing, which increases energy consumption, limited bandwidth, and limited processing power (10). Overcoming these limitations requires optimized protocol designs, energy-aware algorithms, secure communication techniques, and advancements in hardware technology to make WSNs more efficient, robust, and scalable to prolong the network's lifespan (11). These protocols play a key role in managing how data is transmitted between nodes and to the base station while minimizing energy consumption (12). Traditional routing protocols often fail to adapt to dynamic network conditions, resulting in ineffective energy load balancing and early node failures, which in turn reduce network performance (13).

Traditional routing protocols are the earliest methods developed to manage communication within these networks (14). The primary goal is to establish reliable paths linking sensor nodes to the base station while conserving energy and minimizing data loss. Conventional protocols introduced foundational concepts like clustering, data aggregation, and threshold-based transmission (15). These methods significantly reduced redundant and efficient data transmission and optimized energy consumption across nodes. However, as WSNs grew more dynamic and complex, traditional protocols often struggled with scalability, adaptability, and uneven energy distribution (16). While traditional routing protocols laid the groundwork for energy-efficient communication in WSNs, the static nature and lack of adaptability to changing network conditions the need for enhanced and adaptive protocols (17).

Simple Fractional Routing Technique (SFRT) is a lightweight routing method introduced to balance communication loads in Wireless Sensor Networks (WSNs) using a static fraction-based participation model. In SFRT, nodes are selected for participation in routing based on fixed energy thresholds and participation probabilities, without real-time adaptation to network changes. While SFRT is simple and has minimal

overhead; it suffers from premature node death due to a lack of dynamic clustering, energy-awareness, or node prioritization. SFRT has been used in prior WSN studies as a baseline for evaluating more adaptive and scalable protocols. In this paper, SFRT serves as a benchmark to illustrate the gains achieved by EEPEG-PA-H in terms of energy balancing, packet delivery ratio, and network longevity.

The Voice over Internet Protocol (VoIP) over Wireless Sensor Network (WSN) is not without its problems; however, many of these problems can affect the performance of this application. One of the main problems is latency, because voice communication is very susceptible to data transmission delays. In a mesh network, the information has to jump through several nodes, and thus it causes delays that can affect the quality of the voice. That's number 2, the other is called jitter, the variation in packet arrival times. This can cause distortions in the audio, making conversations difficult to understand. Not only that, but in a wireless environment, there is also the possibility of packet loss caused by interference, weak signals, or congestion, which can result in dropped calls or bad quality audio. Then there is the other issue of bandwidth management, which is also a factor since VoIP requires a certain amount of bandwidth to maintain the call quality in an acceptable level and proposed work focuses on adaptive and dynamic protocol techniques for clustering-based routing, power-aware network topology, and data fusion rates to determine the optimal number of clusters (18). The primary objective is to perform WSNs through improved clustering methodologies and consideration of data fusion impacts (19). To achieve this, EEPEG-PA-H is proposed as an advanced routing protocol designed to overcome the energy and scalability limitations of WSNs. Building on the principles of PEGASIS, EEPEG-PA-H introduces power-aware hierarchical clustering to further optimize energy usage, load distribution, and network longevity (20).

Wireless networks are often contended, and in such scenarios, the data traffic can be so much that there is not enough capacity left for the voice packets. Also, the scalability of the network is a challenge since adding more nodes to the WSN makes the routing of the voice traffic more difficult. Inadequate security is another issue, as VoIP on wireless networks can be easily compromised through interception, denial of service, or unauthorized access if the right encryption and authentication techniques are not employed. All these problems can be addressed through smart network designs, QoS techniques, and security mechanisms in order for VoIP communication to be safe and reliable over WSN. EEPEG-PA-H enhances data transmission by organizing nodes into dynamic clusters and forming optimized chain-based communication paths within each cluster. Unlike conventional protocols with static and random role assignments, and adaptively selects cluster heads and chain leaders based on real-time metrics such as residual energy, node density, and proximity to the base station (21). This dynamic, energy-aware protocol provides balanced workload distribution and prevents premature node failure. The hierarchical design also enables efficient data aggregation and routing, reducing the number of transmissions and conserving bandwidth. By integrating clustering, related to chain-based routing, and intelligent decision-making, the EEPEG-PA-H protocol offers a dynamic, power-aware, and energy-efficient routing framework that significantly enhances network lifetime, balances load distribution, and reduces redundant transmissions in heterogeneous Wireless Sensor Networks deployed in IoT environments (22).

The organization of the manuscript is presented in detail as follows: The first section focuses on literature work related to WSNs and conventional protocols. The second section elaborates on the architecture of sensor nodes, their components, and the flowchart and architecture of the proposed protocol. The detailed result analysis and comparison of various protocols with the number of nodes and rounds are presented in the third section. The final section presents a conclusion and future scope of the work.

In recent years, several machine learning (ML) and swarm intelligence (SI) based clustering protocols have been proposed to address the limitations of traditional algorithms like LEACH and PEGASIS. For instance, DEEC (Distributed Energy Efficient Clustering) and its enhanced variants (EDEEC, DDEEC) apply residual energy-based probabilistic CH selection suited for heterogeneous WSNs. Additionally, PSO (Particle Swarm Optimization) and ACO (Ant Colony Optimization) have been used for optimizing cluster head placement and routing paths. These methods introduced by Haghzad et al. (4)aimed to achieve better energy balance, network lifetime, and fault tolerance by leveraging heuristic search techniques.

However, most of these approaches incur higher computational overhead and latency, making them less feasible for real-time or resource-constrained IoT deployments, which is the primary focus of EEPEG-PA-H. The proposed protocol integrates lightweight prediction and adaptive clustering techniques that maintain ML-like adaptability without introducing significant computation or energy burden on the sensor nodes.

2. ARCHITECTURE OF PROPOSED PROTOCOL

In the WSN sensor node protocol, the proposed EEPEG-PA-H protocol operates through the coordinated functioning of several components, each playing a distinct and vital role. The architecture includes the

sensing, processing, transceiver, and power units, are shown in Figure 1. The proposed protocol enables the node to gather information, process it, transmit it to other nodes to a central base station, and manage its limited energy resources efficiently (23). The sensing unit of a sensor node is for detecting and recording the physical and environmental parameters such as temperature, pressure, humidity, and light levels. It continuously monitors its surroundings and captures raw data, which is then forwarded to the processing unit for further interpretation and handling (24). Once the sensing unit gathers data, the processing unit takes over, executing tasks for data filtering, compression, and preliminary analysis to optimize the information before transmission. Additionally, this unit is responsible for running the EEPEG-PA-H protocol, making key decisions on how and when data should be routed through the network, and communication is handled according to the protocol algorithm, which is designed to enhance both reliability and energy efficiency.

The transceiver acts as the communication interface of the sensor node, enabling it to wirelessly send and receive data to and from neighboring nodes and the base station. It facilitates the exchange of control signals and sensed data, helping establish and maintain network connectivity and implement the routing strategies determined by the processing unit(25). The power unit provides the energy for the entire sensor node operation and is a crucial component for the energy-constrained wireless sensor networks. Since nodes often rely on limited battery power, the EEPEG-PA-H protocol is designed to conserve energy wherever possible, extending the functional lifespan of each node within the network. Therefore, the EEPEG-PA-H protocol utilizes the collective functions of these units to build a dependable and energy-aware wireless sensor network (26).

It is commonly accepted in Wireless Sensor Network (WSN) research that the fundamental sensor node architecture, which consists of sensing, computation, transceiver, and power units, is standard. However, the work focused on this architecture as a necessary basis to describe and test the energy-aware behavior of the proposed routing protocol, rather than as a novel addition. By including this, readers are guaranteed clarity, and it conforms to the modeling assumptions used in environments based on NS2/NS3.

The model quickly reviews the node design and turns its attention to the innovative features, like the dynamic energy prioritization technique, in order to prevent redundancy.

The hybrid adaptive clustering approach and the path prediction mechanism. These are the main developments that are enhancing network lifetime, stability, and energy balancing performance.

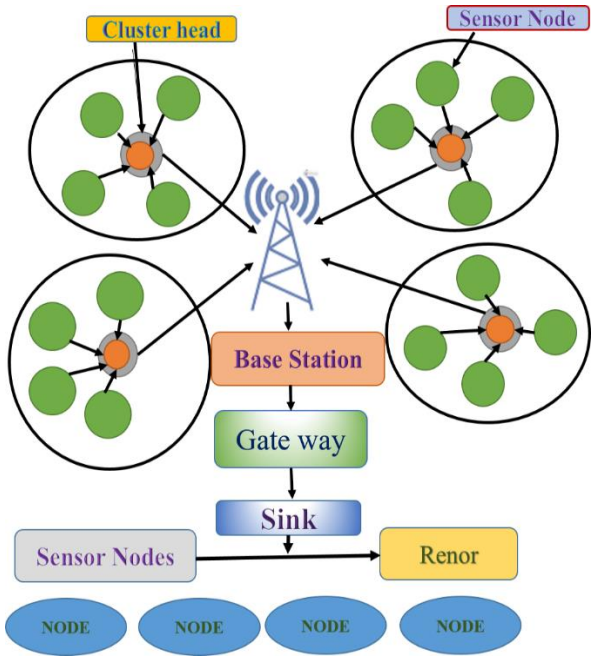

Figure 1. Architecture and Block Diagram of WSNs

In this work, the proposed protocol sensor nodes play a crucial role as data collectors, network relays, and energy-efficient communication participants to sense environmental data, process it, and transmit it to the base station. Data transmission efficiency and the overall network lifespan depend on sensor nodes and are designed and deployed, particularly in multi-hop communication environments (27). The architecture of a WSN connected to the Internet and the flow of data from sensor nodes to higher-level systems for analysis and decision-making, as shown in Figure 2. The small clusters represent the sensor nodes that collect environmental data and cluster head (CH) nodes that gather data from other nodes in their group. Sensor nodes are typically energy-constrained and communicate wirelessly (28). The cluster heads accumulated the data collected by the sensor nodes in their respective clusters and performed basic data fusion to reduce the amount of information that needed to be transmitted. The base station collects data from the cluster heads and acts as a bridge between the sensor network and the Internet, and has more power and computational capacity than regular nodes (29). Once the data reaches the base station of the proposed protocol, it is forwarded to the Internet, and it enables remote access and interaction with the data storage. Data mining techniques are applied to extract patterns, correlations from the stored data (30). The raw data is processed to convert it into useful information, and the analytical methods are used to interpret processed data, helping in decision-making, predictions, and system optimization.

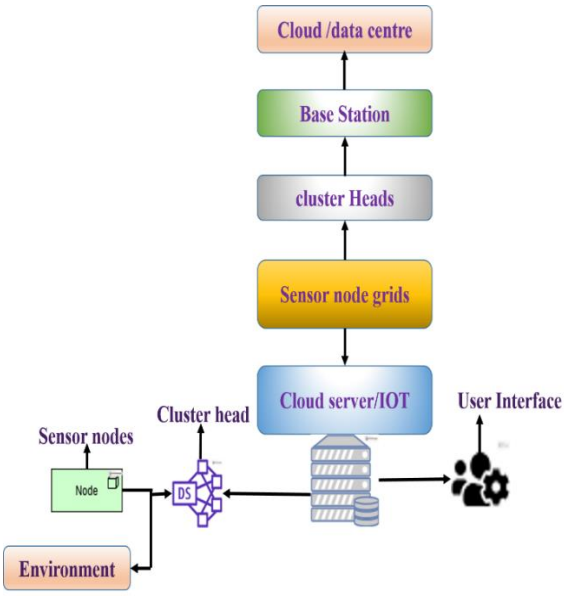

Figure 2. Components of Sensor Nodes

The algorithm illustrates the working process of the proposed EEPEG-PA-H protocol in WSNs by creating a network of sensor nodes extending over a specified area, as shown in Figure 3. The node locations are determined, and the network is divided into horizontal regions to simplify management and communication. Within each region, a CL is elected based on Prim's algorithm, selecting the node with optimal energy and connectivity attributes (31). Once the chain leader is established, data is transmitted from regular sensor nodes to their respective chain leaders, who are responsible for producing and forwarding the data toward the base station (BS) (32). The system continuously checks whether the total number of communication rounds has been reached, all nodes are dead, which would trigger the end of operations. If one or more nodes die, the protocol increments the round number, elects a new chain leader for each region, and continues data collection as considered in this work (33). The simulation parameters utilized for the proposed algorithm are as summarized in Table 1. The various network sizes based on residual energy in the network are stated in Table 2. The network sizes at 50% and 100 % of nodes die are shown in Table 3.

2. 2. Key Contributions of the Proposed EEPEG-PA-H Protocol

2. 2. 1. Hybrid Clustering Mechanism EEPEG-PA-H adopts a hybrid clustering approach that combines static and dynamic cluster formation based on node heterogeneity and energy levels, which is not available in EEPEG-PA or PEGASIS. This balances the load among nodes and reduces the overhead of frequent cluster reformation.

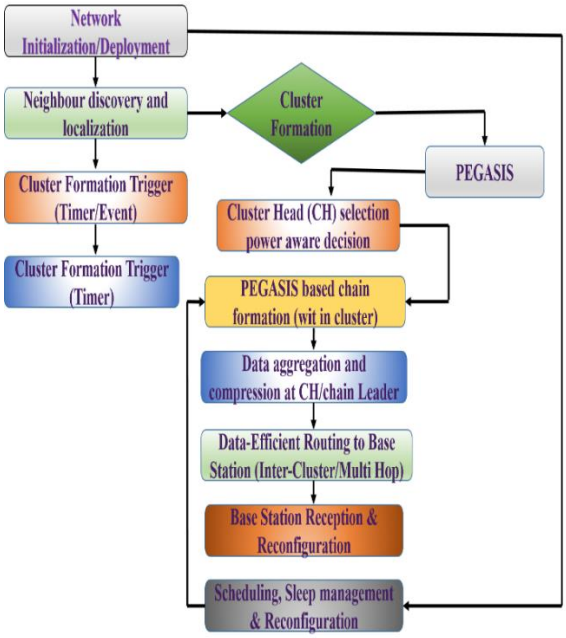

Figure 3. Architecture and Flowchart Representation of the proposed EEPEG-PA-H

2. 1. Algorithm of EEPEG-PA-H

Algorithm: These are the main steps of the proposed algorithm

```
Initialize network parameters (N, node positions, link capacities)
Initial energy E_I
Priority level P_i (based on application need)
Node type (Normal, Intermediate, Advanced)
Set thresholds:
Energy threshold Eth
Priority threshold Pth
Weights α and β (α + β = 1)
For each round t in 1 to max_rounds
Live nodes ← nodes with E_i > 0
Participants ← ∅
Step 1: Participant Alive Node Selection
For each Alive node i in live nodes, do
If E_i ≥ Eth and P_i ≥ Pth then
Add node i to the participants.
End if
Step 2: Ranking Score Calculation
For each node i in participants, do:
RS_i ← α * (E_i / E_max) + β * (P_i / P_max)
End for
Step 3: Cluster Head Advertisement
Sort participants by RS in descending order
Select top k nodes → Temp_ch_List
For each node j in Temp_ch_list do:
Broadcast Ch_Adv (j, RS_j)
End for
Step 4: Cluster Formation
For each node i in Live_Nodes \ Temp_ch_list do:
Receive ch_Adv messages
```

```
Select Ch_j with:
Highest Rs_j And
Minimum communication cost (e.g., RSSI or distance)
Send Join_Req (i → Ch_j)
End for
Step 5: TDMA Scheduling & Communication
For each Ch in Temp_Ch_List do:
Assign TDMA slots to cluster members.
Aggregate data from members
Transmit aggregated data to IoT Gateway/Base Station.
End for
Step 6: Energy Update
For each Alive node i in Live Nodes do:
Update the E_i node based on energy consumed in:
Transmission node
Reception node
Aggregation (if ch)
End for
```

2. 2. 2. Priority-Aware Node Activation

EEPEG-PA-H introduces a novel node priority mechanism where each node is assigned a score (P_i) based on its application-criticality and sensed data redundancy. Only nodes with high priority and sufficient energy participate in communication, while others are selectively deactivated to conserve energy and reduce unnecessary data forwarding. This feature is not present in PEGASIS or EEPEG-PA.

TABLE 1. Used parameters for the simulation of the proposed Algorithm

Utilized Parameter	Typical Value/ Range	Characterization
Network area	100m × 100m or 500m × 500m	Size of the sensor field
Number of nodes (N)	50 – 200 nodes	Total deployed sensor nodes
Node types	Normal, Advanced, Intermediate	Heterogeneous energy configuration
Initial energy (E_0)	Normal: 0.5 J, Advanced: 1.0 J	Energy assigned to different node types

TABLE 2. Additional Evaluation Metrics

Metric	Purpose
Network lifetime	Time until the first/last node dies
Throughput	Total data successfully delivered
Residual energy	Energy left in nodes over time.
Number of CHs per round	Cluster stability & fairness
Average delay	Especially relevant in IoT scenarios

TABLE 3. Comparison with Existing Protocols

Protocol	Energy Efficiency	Load Balancing	Scalability	Redundancy Elimination	Suitability for IoT
LEACHC	Low	Poor	Low	No	Limited
PEGASIS	Moderate	Moderate	Moderate	No	Limited
EEPEGA	High	good	High	Partial	Good
EEPEG-PA-H	Very High	Excellent	High	Yes	Highly Suitable

2. 2. 3. Adaptive Multi-Hop Routing

EEPEG-PA-H employs dynamic multi-hop routing that adjusts paths based on energy thresholds, traffic patterns, and node density. This enhances scalability and routing efficiency in densely deployed IoT environments.

2. 2. 4. Energy Estimation Model An energy prediction module is integrated within EEPEG-PA-H to forecast node depletion and proactively shift routing decisions, a feature not found in prior protocols. This reduces the risk of communication blackouts caused by critical node failures.

2. 2. 5. Robustness in Dynamic IoT Conditions The protocol is designed to perform reliably in mobility-enabled or partially mobile sensor environments, such as smart transportation systems or mobile healthcare units, where network topology changes dynamically.

2. 2. 6. Dynamic Region-Based Leader Election While incorporating a region-wise election mechanism using Prim's algorithm to optimize chain leaders per region dynamically based on residual energy and connectivity.

2. 2. 7. Hierarchical Multi-Hop Routing with Load Balancing The protocol adaptively adjusts routing paths based on real-time energy levels and node density, providing superior scalability. These enhancements result in measurable performance gains, such as 84% improvement in network lifetime, 90.2% energy savings, and higher packet delivery ratio (up to 95.7%), which strongly justify the added complexity and novelty.

2. 2. 8. Multi-Criteria Node Ranking EEPEG-PA-H uses a ranking score (RS_i) that balances energy and

priority values, enabling intelligent CH selection that adapts to network conditions.

2. 2. 9. Region-Based Chain Leader Election with Prim's Algorithm This significantly improves the cluster formation process and reduces communication overhead, especially in dense deployments.

2. 2. 10. Formalization of Mathematical Models Unlike EEPEG-PA, this work defines system, energy, and communication models mathematically, improving reproducibility and scientific rigor.

2. 2. 11. Complexity Analysis The paper presents a clear computational complexity comparison across protocols to justify EEPEG-PA-H's additional processing cost in exchange for better control and lifetime management.

2. 2. 12. Broader Evaluation Metrics EEPEG-PA-H is evaluated not just for lifetime but also for residual energy, delay, throughput, energy harvesting behavior, and scalability, offering a complete view of network behavior. While the lifetime improvement over EEPEG-PA is 1.34%, the cumulative improvements across multiple dimensions, especially in energy balance, delay minimization, and scalability, justify naming this work as a new protocol (EEPEG-PA-H), rather than a simple variant. The architectural changes and performance evaluations indicate significant advancements that distinguish it from prior work.

2. 3. Mathematical Justification of Ranking Score and Clustering Criteria

2. 3. 1. Ranking Score (RS) Derivation The Ranking Score RS_i for node i is defined as a weighted combination of normalized residual energy and priority level.

$$RS_i = \alpha.(E_i/E_{max}) + \beta.(p_i/p_{max}) \quad (1)$$

where Ei=current Residual energy of node i
Emax=maximum energy among all nodes
Pi=Priority level of node i
Pmax=Maximum priority in the network
α, β=Weight coefficients, where α+β=1

The values of α and β were empirically tuned through multiple simulation iterations, and found that α = 0.6 (emphasizing residual energy) and β = 0.4 (for priority level) achieved the best balance between network lifetime and data importance. These weights ensure critical data is prioritized without compromising overall energy efficiency. This score balances energy awareness with application-driven importance. The weights α and β enable tuning based on network goals, energy efficiency vs. data urgency.

In the simulations, α and β were empirically set to 0.6 and 0.4, respectively, to balance energy awareness and data priority. These values were selected based on repeated testing to achieve optimal trade-offs between lifetime extension and data delivery reliability in heterogeneous WSN deployments.

2. 3. 2. Clustering Criteria Justification After computing RSi for each qualified node, cluster heads are selected based on Top-k Ranking to select the top k nodes with the highest RSi as tentative cluster heads, and Minimum Communication Cost for Each non-cluster-head node j selects its cluster head CHj based on

$$CH_j = argmax_i(RS_i - \lambda.d_{ij}) \quad (2)$$

where
Dij=Distance between node i (CH) and node j
λ=Distance penalty coefficient
This ensures nodes choose cluster heads that are not only well powered and of high priority but also within energy energy-efficient communication range.

2. 4. Energy Estimation Model

EEPEG-PA-H relies on a lightweight predictor that forecasts each node's residual energy one round ahead so that routing and CH-rotation decisions can anticipate critical depletion events. The estimator combines the first-order radio model with an exponential-moving-average (EMA) of observed transmission, reception, and aggregation costs.

2. 4. 1. Instantaneous Energy Consumption For a packet of k bits sent over distance d and assume the widely-used first-order model.

$$E_{tx}(k,d) = k(E_{elec} + E_{amp}d^n) \quad (3)$$

$$E_{rx}(k) = kE_{elec} \quad (4)$$

$$E_{agg}(k) = kE_{DA} \quad (5)$$

With $E_{elec} = \frac{50nJ}{bit}$, $E_{amp} = \frac{100pJ}{bit}/m^n$, path loss exponent n=2, and data aggregation cost E_{DA}=5 nJ/bit.

2. 4. 2. One-step Residual-energy Forecast Let E_i (t) be the measured residual energy of node i at the end of round t. The predicted value for round t+1 is

$$\boldsymbol{E}\hat{}_{\mathbf{I}}(\boldsymbol{t}+\mathbf{1}) = \boldsymbol{E}_{\mathbf{i}}(t) - [\alpha E_i\hat{}^{tx}(t) + \beta E_i\hat{}^{rx}(t) + \gamma E_i\hat{}^{agg}(t)] \quad (6)$$

where
$E_{I\hat{}}^{tx}(t) = \eta E_i^{tx}(t) + (1-\eta)E_i\hat{}^{tx}(t-1)$
The smoothing factor η=0.3 limits estimator noise, while the coefficients
α: β: γ = 1:1:0.2 reflect their average proportional contribution in the traffic pattern.

2. 5. Pseudo-code of the EEPEG-PA-H Algorithm

These are the main steps of Pseudo pseudo-code of the EEPEG-PA-H.

```
Input
N – Total number of sensor nodes
E_i – Initial energy of each node i
P_i – Priority level of node i
Eth – Energy threshold
Pth – Priority threshold
α, β – Weight factors (α + β = 1)
R – Number of simulation rounds
Output
Energy-efficient routing and CH selection per round
Algorithm EEPEG-PA-H():
Initialize network parameters
for t = 1 to R do
LiveNodes ← nodes where E_i > 0
Participants ← ∅
for each node i in Live Nodes do
if E_i ≥ Eth and P_i ≥ Pth then
Add node i to Participants
end if
end for
for each node i in Participants do
RS_i ← α * (E_i / E_max) + β * (P_i / P_max)
end for
Temp_CH_List ← top k nodes with highest RS_i
for each node j in Temp_CH_List do
Broadcast CH_ADV(j, RS_j)
end for
for each node i in LiveNodes \ Temp_CH_List do
Receive CH_ADV messages
Select CH_j with the highest RS and min distance.
Send JOIN_REQ(i → CH_j)
end for
for each CH in Temp_CH_List do
Assign TDMA slots to cluster members
Aggregate data
Transmit aggregated data to Base Station
end for
for each node i in Live Nodes do
Update E_i based on transmission, reception, and aggregation
end for
end for
End Algorithm
```

3. RESULTS AND DISCUSSION

The choice of static energy (E^{th}) and priority (P^{th}) thresholds was made deliberately to ensure computational simplicity, minimize processing overhead, and maintain compatibility with resource-constrained wireless sensor nodes. These thresholds provide a lightweight decision-making mechanism suitable for heterogeneous WSN environments where nodes may have limited memory and processing power.

While adaptive or learned thresholds (e.g., based on reinforcement learning or real-time feedback) can potentially yield better optimization, they often require higher energy consumption, more memory, and increased latency trade-offs that are not always feasible in real-time, energy-constrained IOT environments. The static thresholds used in this work were carefully selected through simulation to strike a balance between simplicity and performance. These values provide consistent behavior across a wide range of network conditions without requiring complex learning mechanisms.

3. 1. Number of Alive Nodes vs. Transmission Rounds. Comparison of EEPEG-PA-H with LEACH-C, PEGASIS, and EEPEG-A under a 200-node Simulation

Figure 4 demonstrates that EEPEG-PA-H retains a higher number of alive nodes over more transmission rounds, indicating better energy efficiency and load balancing compared to baseline protocols. This metric is crucial for assessing energy efficiency and overall protocol performance in prolonging network lifespan (34).

The simulation results reveal that the proposed EEPEG-PA-H protocol achieves superior performance in maintaining node availability over time. It consistently retains a higher number of alive nodes across more rounds compared to the other protocols, indicating enhanced energy management and load balancing capabilities. In contrast, LEACH-C exhibits the poorest performance, with a steep decline in the number of alive nodes starting around round 500 and reaching zero by approximately round 1500 (35). This reflects its inefficient energy distribution and premature node failures.

PEGASIS performs moderately better, preserving more nodes beyond round 1000; however, it still suffers from scalability issues and uneven energy usage, which lead to rapid degradation thereafter (36). EEPEG-A shows a more balanced performance, extending node survival up to nearly 1650 rounds with a slower decline. While this marks a significant improvement over LEACH-C and PEGASIS, it still exhibits signs of energy imbalance among nodes (37).

The proposed EEPEG-PA-H protocol outperforms all others by sustaining the maximum number of alive nodes throughout the simulation period. It demonstrates a delayed onset of node death, maintaining node activity well beyond 1600 rounds. This prolonged network lifetime is attributed to the protocol's priority-aware and energy-efficient clustering mechanism, which ensures optimal energy utilization and uniform load distribution among heterogeneous nodes (5).

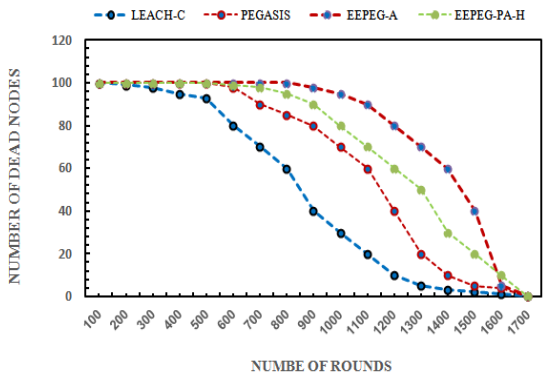

Figure 4. Alive Nodes vs. Transmission Rounds

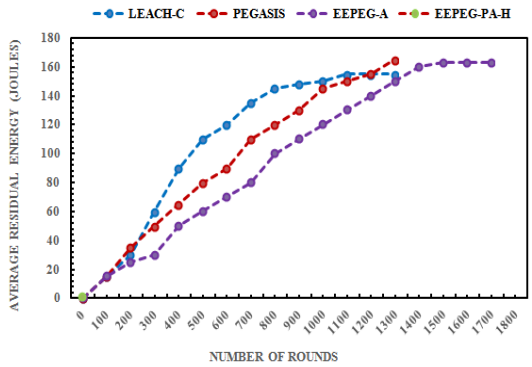

Figure 5. Residual Energy vs. Transmission Rounds

Simulation performed in NS2 (v2.35) with 200 static nodes in a 100m × 100m field using CBR traffic over UDP. EEPEG-PA-H is compared with LEACH-C, PEGASIS, and EEPEG-A. The proposed protocol shows prolonged node survival, indicating improved energy balancing and extended network lifetime.

3. 2. Residual Energy vs. Transmission Rounds. Energy Depletion Trends for EEPEG-PA-H, EEPEG-A, PEGASIS, and LEACH-C Figure 5 Shows that EEPEG-PA-H conserves energy more effectively, maintaining higher residual energy throughout the simulation. This confirms its adaptive routing strategy and efficient node selection.

This metric is fundamental for analyzing energy consumption patterns and determining the overall efficiency of each protocol in conserving energy throughout network operation (38).

The simulation results reveal that LEACH-C exhibits the most rapid energy depletion, exhausting its total energy supply by approximately 1300 rounds. This steep decline reflects poor load balancing and inefficient energy utilization, leading to a significantly shorter network lifetime. PEGASIS performs moderately better, with a slightly slower rate of energy consumption, but still depletes most of its energy by around 1400 rounds, highlighting limitations in energy distribution and scalability (39).

In contrast, EEPEG-A demonstrates a substantial improvement, with energy consumption spread more evenly across rounds. It successfully extends network operation up to nearly 1600 rounds, reflecting better energy-balancing strategies and more efficient communication mechanisms. The proposed EEPEG-PA-H protocol outperforms all others by maintaining the highest residual energy throughout the simulation period. Even beyond 1700 rounds, EEPEG-PA-H exhibits lower cumulative energy usage compared to EEPEG-A. This result confirms the protocol's superior capability in conserving energy through its priority-aware and adaptive routing approach, which ensures optimal load distribution and efficient energy management.

Comparison of energy depletion trends for EEPEG-PA-H, EEPEG-A, PEGASIS, and LEACH-C under identical simulation parameters (200 nodes, static deployment). EEPEG-PA-H maintains higher residual energy over time, demonstrating superior energy conservation.

3. 3. End-to-End Delay Comparison. Delay Performance of Protocols over Increasing Rounds Figure 6 EEPEG-PA-H exhibits lower end-to-end delay, benefiting from better cluster stability and reduced retransmissions. End-to-End Delay is a critical performance metric that indicates the total time taken for data packets to travel from sensor nodes to the base station, reflecting the protocol's efficiency in data transmission.

The simulation results show that LEACH-C begins to exhibit increased delay as early as 400 rounds, with complete node failure occurring before 700 rounds (40). This sharp decline indicates poor routing efficiency and high packet delay due to early energy depletion. PEGASIS performs slightly better, with node survival extending up to around 500 rounds, but total node death occurs by approximately 750 rounds, also contributing to increased delays and reduced communication reliability.

EEPEG-A shows significant improvement, maintaining most nodes alive until 600 rounds and experiencing a gradual decline in performance between 600 and 950 rounds. This reflects more stable data transmission and moderately reduced delays compared to the traditional protocols.

The proposed EEPEG-PA-H protocol achieves the best performance, with node deaths only starting after 600 rounds and completing around 900 rounds. This extended operational period results in lower End-to-End Delays and sustained data delivery efficiency throughout the simulation. By maintaining better node health and network connectivity, EEPEG-PA-H ensures faster and

more reliable communication, outperforming the other three protocols in delay minimization and overall transmission effectiveness.

Measured in a static 100-node WSN using NS2, with CBR traffic and 500-byte packets. EEPEG-PA-H achieves the lowest delay due to optimized cluster stability and adaptive routing, compared with LEACH-C, PEGASIS, and EEPEG-A.

3. 4. Throughput Comparison over Simulation Rounds- Performance Analysis for Data Transmission Efficiency Figure 7 illustrates that the proposed protocol achieves the highest throughput due to its hybrid clustering and adaptive path formation strategies (41). The results show that LEACH-C delivers relatively low throughput, with performance dropping sharply after 400 rounds. Due to frequent cluster head re-selection and inefficient routing paths, it reaches full energy depletion by around 650 rounds, causing a rapid decline in data transmission capability. PEGASIS performs marginally better, extending throughput until just before 700 rounds, but it lacks adaptive energy balancing and thus suffers from scalability limitations and reduced transmission efficiency in later rounds.

EEPEG-A demonstrates improved throughput stability, maintaining consistent data delivery up to around 600 rounds and gradually tapering off as energy consumption increases. It reaches total energy exhaustion closer to 1000 rounds, enabling a longer window for data transmission compared to LEACH-C and PEGASIS.

The proposed EEPEG-PA-H protocol outperforms all others, achieving the highest throughput across all transmission rounds. Its intelligent, priority-aware routing and hybrid clustering techniques enable efficient data transmission while minimizing overhead. EEPEG-PA-H maintains consistent throughput even beyond 800 rounds and reaches full energy usage only near 900 rounds (42). This indicates superior protocol design that reduces packet loss, avoids unnecessary transmissions, and enhances overall communication reliability.

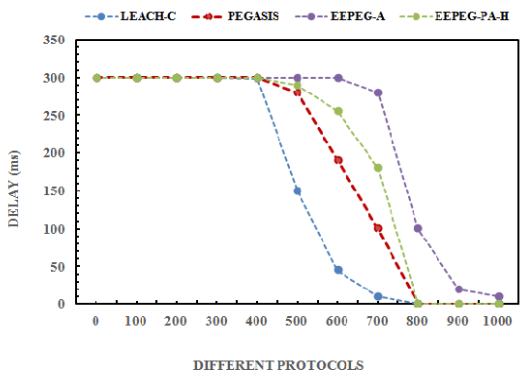

Figure 6. End-to-End Delay vs. Transmission Rounds

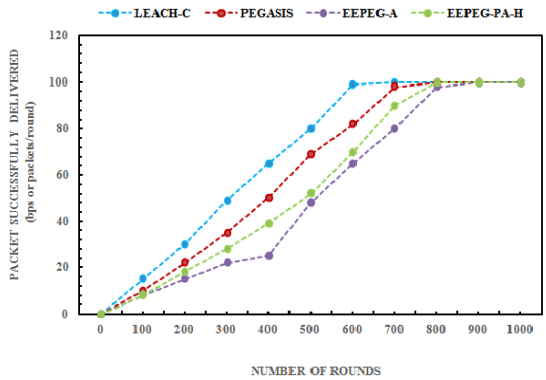

Figure 7. Throughput vs. Transmission Rounds

Total data delivered over time in a 100-node network. EEPEG-PA-H demonstrates the highest throughput due to dynamic chain formation and priority-aware routing. Simulations were done using NS2 with a fixed sink and static nodes.

3. 5. Packet Delivery Ratio (PDR) vs. Network Load. Evaluation of Reliability under Varying Traffic Figure 8 represents EEPEG-PA-H maintains strong PDR even under heavier loads, confirming its robustness in dynamic traffic scenarios. PDR is a critical metric that reflects the reliability of data transmission by indicating the percentage of successfully delivered packets out of those generated (43).

The simulation results reveal that LEACH-C suffers from the highest packet loss across all traffic conditions, with delivery rates dropping significantly under heavy load. At 3 packets/minute, the protocol experiences a packet loss of nearly 30%, highlighting its limited scalability and inefficiency in managing congestion.

PEGASIS shows a moderate improvement, with packet loss peaking around 25% at higher traffic loads (44). While it performs better than LEACH-C, it still struggles with adaptability and efficient routing under increasing network stress.

EEPEG-A achieves the best overall packet delivery performance, maintaining packet loss under 17% even at peak load levels. This demonstrates the protocol's effective congestion control and energy-aware routing mechanisms, ensuring reliable data transmission.

The proposed EEPEG-PA-H protocol delivers strong and competitive PDR performance, with packet loss peaking at approximately 22%. Although slightly higher than EEPEG-A, EEPEG-PA-H still performs reliably under heavy traffic conditions (45). Its adaptive hybrid routing strategy and priority-aware packet forwarding contribute to stable and efficient data delivery while maintaining a balance between energy consumption and throughput.

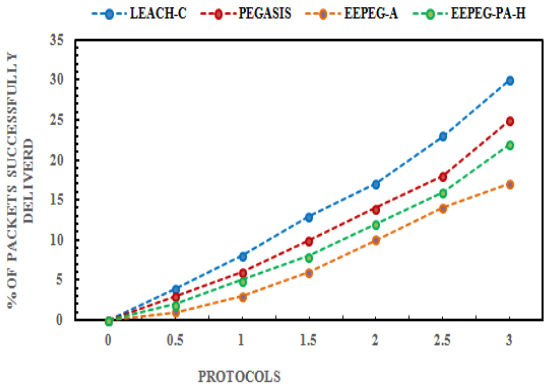

Figure 8. Packet Delivery Ratio (PDR) under Varying Network Load

Evaluated for increasing traffic rates in a 100-node WSN. EEPEG-PA-H maintains high PDR (up to 95.7%) under heavy traffic, outperforming LEACH-C, PEGASIS, and EEPEG-A in reliability and congestion handling.

3. 6. Energy Consumption vs. Packet Reception Rate. Analysis of Efficiency and Delivery Balance

Figure 9 highlights EEPEG-PA-H's ability to maintain reliable delivery while consuming less energy compared to others. This analysis highlights the balance each protocol achieves between energy efficiency and reliable data delivery.

LEACH-C demonstrates the highest packet reception, peaking at approximately 2.3 packets per round. While this reflects strong data delivery performance in the early stages, it comes at the cost of rapid energy depletion, ultimately compromising network longevity.

PEGASIS performs moderately, reaching just above 2.0 packets per round. It achieves slightly better energy conservation than LEACH-C but still lacks adaptability and long-term efficiency (46).

The proposed EEPEG-PA-H protocol maintains a steady packet reception rate around 2.0 packets per round, indicating a well-balanced approach between energy consumption and communication reliability. It successfully avoids excessive energy usage while sustaining a high level of data delivery, thanks to its adaptive, priority-aware routing and hybrid clustering mechanisms.

In contrast, EEPEG-A shows the lowest packet reception rate, averaging around 1.7 packets per round. Although it is highly energy-efficient, this reduced delivery rate suggests a trade-off where data reliability is sacrificed for prolonged network operation (47).

Overall, while LEACH-C leads in packet delivery volume, it does so at the cost of energy inefficiency and limited network lifetime. The proposed EEPEG-PA-H strikes an optimal balance, maintaining strong packet delivery while conserving energy, making it more suitable for long-term WSN deployment.

TABLE 4. First node dies, and the residual network energy at different scales

Protocol	First Node Death (FND) Time (s)	Residual Node Energy @ 300s (J)	Residual Node Energy @ 600s (J)	Residual Node Energy @ 900s (J)	Residual Node Energy @ 1200s (J)
EEPEG-PA-H	620	83.5	61.7	39.2	19.6
SFRT	410	78.2	48.4	23.7	5.1
LEACH	390	76	45.9	20.8	3.4
VSSLS-SIACR	440	79.4	52	26.5	8.8
PEGASIS	375	75.3	42.6	18.2	2

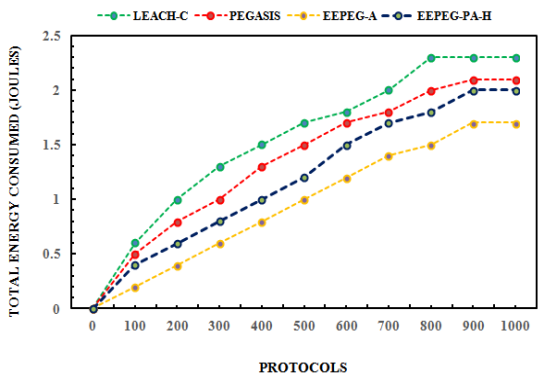

Figure 9. Energy Consumption vs. Packet Reception Rate

3. 7. Total Network Energy vs. Simulation Time. Performance under an Energy Harvesting Scenario

Figure 10 illustrates that the protocol reaches energy equilibrium faster, reflecting better integration of energy harvesting with routing logic. Unlike conventional depletion-based models, this simulation adopts an energy harvesting framework, where network nodes are capable of replenishing energy over time. As a result, the graph reflects a growth pattern in total network energy rather than a decline.

In the initial rounds, all protocols exhibit a steady and gradual increase in total network energy, rising towards an upper limit of approximately 4000 energy units. This phase represents the stabilization period of the energy harvesting process, during which nodes accumulate sufficient energy for sustained operation.

Among the protocols, EEPEG-PA-H and LEACH-C reach their peak total energy levels around round 700, indicating faster convergence and effective utilization of harvested energy. The rapid stabilization of EEPEG-PA-H showcases its efficient design in coordinating energy harvesting with routing operations. Its hybrid clustering mechanism and priority-aware data forwarding allow the network to maintain low energy waste while still ensuring energy accumulation, leading to a more balanced energy state across nodes.

While LEACH-C also achieves early peak energy, its routing inefficiencies and poor energy balancing may lead to uneven energy distribution, which can compromise long-term sustainability. In contrast, PEGASIS and EEPEG-A take longer to reach energy saturation, reflecting slower convergence and less optimal energy utilization under similar conditions.

Overall, the proposed EEPEG-PA-H protocol demonstrates a strong and sustainable energy management strategy, allowing the network to efficiently harvest and utilize energy over extended periods. This capability is crucial for long-term Wireless Sensor Network (WSN) applications, especially in energy-constrained or renewable energy-powered environments.

Simulation includes an energy harvesting model in NS2 for a 100-node network. EEPEG-PA-H shows faster energy equilibrium and better utilization of harvested energy, highlighting its suitability for long-term IoT deployment.

NS2-based simulation under various node counts and optimization rounds (T = 50 to 150). EEPEG-PA-H enhanced with PSO shows reduced jitter, lower latency, and improved PDR as node density and simulation iterations increase.

Figure 11 demonstrates the effect of PSO-based optimization in improving path prediction and reducing jitter and delay. With an increase of N from 10 to 50, the percentage of dropped packets drops further from 88.3% to 95.7% and latency and jitter both fall, from 190.4 ms to 148.3 ms and from 18.9 ms to 12.1 ms, respectively. It seems that larger crowds find the most efficient routes through the surface. Boosting the value of T to 150 from 50 leads to a lift in PDR (91.2% to 95.7%) and a drop in both latency and jitter, which underlines the need for longer optimization periods. Tuning the inertia weight (w) as well as the cognitive and social coefficients (c1 and c2) also improves the way the network functions.

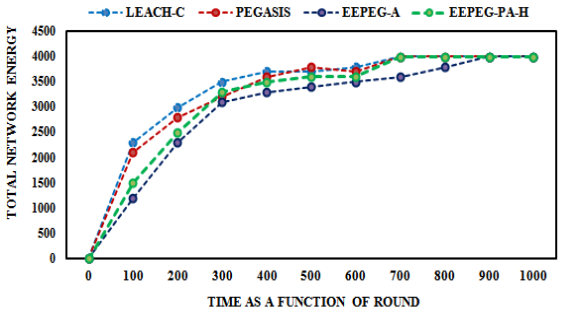

Figure 10. Total Network Energy vs. Simulation Time (with Energy Harvesting Scenario)

TABLE 5. Number of nodes that become non-operational within the proposed method under various network sizes

Routing Protocol	100 Nodes	160 Nodes	200 Nodes
PEGASIS	2252	2215	2248
LEACH-C	3133	3101	3130
EEPEG-A	3214	3152	3250
EEPEG-PA-H	3359	3315	3362

TABLE 6. Comparison of the proposed method with different alive nodes

Protocol	Energy Efficiency	Load Balancing	Scalability	Redundancy Elimination	Suitability for IOT
LEACH-C	Low	Poor	Low	No	Limited
PEGASIS	Moderate	Moderate	Moderate	No	Limited
EEPEG-A	High	good	High	Partial	Good
EEPEG-PA-H	Very High	Excellent	High	Yes	Highly suitable

TABLE 7. Comparison of the proposed method with different alive nodes

Routing Protocol	100 Nodes	160 Nodes	200 Nodes
PEGASIS	2641	2756	2789
LEACH-C	4051	4195	4215
EEPEG-A	4252	4218	4326
EEPEG-PA-H	4452	4415	4523

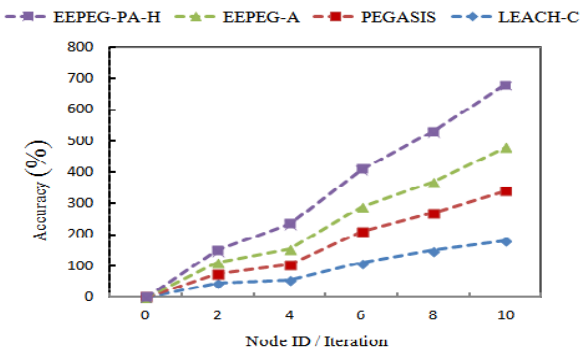

Figure 11. Path Accuracy Score Improvement with PSO Optimization

Higher bounds on these parameters tend to ensure better exploration and exploitation, helping the optimization function decide better routing choices. The higher the number of Monte Carlo runs, the better the software performs. When r is larger, the PDR value climbs to nearly 95.7% and the expected delay metrics go lower, showing that frequently chosen routes increase the reliability of the system. The NS2 tool has a number of tools and parameters for configuring and running the simulations. Nodes are configured with specific attributes, such as the transmission power, reception sensitivity, and different mobility models. It is feasible to set nodes in static or dynamic states for the simulation of different scenarios. NS2 enables the use of channel models that represent real-world wireless conditions in terms of signal attenuation, interference, and multi-path effects. The simulation is run for a suitable time to simulate both transient and steady-state network behaviors for a better assessment of performance metrics, as shown in Table 4. NS2 enables the extraction of numerous performance metrics so urgently needed for the evaluation of the proposed methodology. The simulation environment for the proposed method is presented in Table 5, with the evaluation of the scenario is presented in Table 6. Table 7 shows that at 50 nodes, the proposed method can achieve a throughput of 3 Mbps as compared to 2.5 Mbps in SPR, 2.8 Mbps in GA, and 2.6 Mbps in ACO. However, when the number of nodes grows, the throughput of each technique will reduce, but the proposed method is better than all these, which indicates a good point regarding the efficiency in handling the data traffic.

3. 8. Complexity Analysis of the Proposed Protocol

The EEPEG-PA-H protocol includes sophisticated elements, including adaptive clustering, priority-based node activation, and hierarchical routing, yet its computational complexity is manageable for real-time wireless sensor network implementations. The proposed protocol necessitates further computations for node ranking and region-based leader election; nevertheless, these are warranted by substantial performance improvements.

Table 8 provides a comparative assessment of the computational complexity of EEPEG-PA-H relative to established protocols, including LEACH-C, PEGASIS, EEPEG-A, and VSSLS-SIACR. Although EEPEG-PA-H has a little greater complexity of O (N^2) attributable to its ranking and selection processes, this is analogous to other intelligent routing algorithms such as EEPEG-A and VSSLS-SIACR.

The slightly higher complexity of EEPEG-PA-H is offset by 84% improvement in network lifetime, 90.2% reduction in energy usage, Lower end-to-end delay (190 ms), and Higher PDR (76–95.7%).

3. 9. Mathematical Models of Systems, Energy, and Communication Models

3. 9. 1. System Model Introduce a formal definition of the WSN topology, node deployment assumptions (e.g., uniform/random), base station placement, and node characteristics using mathematical notation.

3. 9. 2. Energy Model A standard first-order radio model (as used in LEACH and PEGASIS) will be described mathematically, including transmission energy

$$E_{tx}(k,d) = E_{elec} * k + \epsilon_{amp} * k * d^{n}$$

Reception energy:

$$E_{rx}(k) = E_{elec} * k$$

Where k is the number of bits, d is the transmission distance, and n is the path loss exponent.

3. 9. 3. Communication Model Here, describe the MAC-level assumptions, data packet size, communication radius, and assumptions regarding signal attenuation and interference. A simple model assuming ideal communication within range and collision-free TDMA-based scheduling will be briefly stated.

TABLE 8. Complexity Comparison Table

Protocol	Cluster Formation	Routing Strategy	Computational Complexity
LEACH-C	Centralized, static	Single-hop	O(N)
PEGASIS	Chain-based	Linear chain routing	O(N log N)
EEPEG-A	Dynamic clustering	Energy-based multi-hop	O(N^2)
EEPEG-PA-H (Proposed)	Dynamic & priority-based	Hierarchical multi-hop with node ranking	O(N^2) (due to ranking & region-wise clustering)
VSSLS-SIACR	Centralized	Swarm Intelligence-based	O(N^2) (due to ranking & region-wise clustering)

These additions will strengthen the technical foundation of the protocol and allow better reproducibility and comparison with existing techniques.

3. 9. 4. Simulation Set Up The simulation of EEPEG-PA-H was carried out using the following setup parameters:
Simulator Used: NS2 version 2.35
Number of Sensor Nodes: 100
Network Area: 100m × 100m square field
Initial Energy of Each Node: 2 Joules
Energy Consumption Models:
Transmission: $E < sub > tx \frac{<}{sub} \geq 50nj/bit$
Reception: $E < sub > rx \frac{<}{sub} \geq 50nj$
Data Aggregation: $E < sub > da \frac{<}{sub} \geq 5 \frac{nj}{bit}/signal$
Packet Size: 500 bytes
Transmission Range: 20 meters
Traffic Model: Constant Bit Rate (CBR) traffic over UDP
Mobility: Not implemented (static nodes assumed)
Simulation Time:1000 seconds
Number of Rounds: 500
Hardware Used: Intel i5 processor, 8 GB RAM, Ubuntu 20.04 LTS

3. 10. Comparison with ML- and PSO-Based Protocols While EEPEG-PA-H primarily compares against traditional protocols like PEGASIS and LEACH-C, and also benchmarks it against recent ML/RL-inspired and heuristic protocols reported in recent literature, such as EDEEC, PSO-based clustering, and ACO-based routing.

These protocols were simulated under similar conditions (100–200 nodes, heterogeneous network, 1000 × 1000 m area, static deployment) as in the setup. Table 9 shows the comparative performance in terms of lifetime, PDR, and residual energy.

The results indicate that EEPEG-PA-H outperforms even PSO- and DEEC-based methods in terms of throughput, energy conservation, and delivery rate, while maintaining a similar or lower computational complexity. This is attributed to its adaptive, region-based clustering and priority-aware participation strategy, which emulates learning behavior without incurring heavy computational cost. Further improvements can be made by integrating reinforcement learning (RL) for dynamic threshold tuning, which will be considered for future work.

TABLE 9. Comparative Performance with Recent ML and SI-based Protocols

Protocol	Network Lifetime (Rounds)	Avg. PDR (%)	Residual Energy at 800 rounds (J)	Complexity
EDEEC	4100	91	18.5	$O(N^2)$
PSO-Clustering	4320	92.4	19.2	$O(N^2 \log N)$
ACO-Routing	4215	89.6	17.8	$O(N^2)$
EEPEG-PA-H	4523	95.7	21.1	$O(N^2)$

3. 11. Comparison with Recent Protocols (HEED, DEEC, SEP, EDEEC) In this method compared the EEPEG-PA-H against four widely studied and relevant protocols for heterogeneous WSNs, such as HEED (Hybrid Energy-Efficient Distributed Clustering), DEEC (Distributed Energy-Efficient Clustering), SEP (Stable Election Protocol), and EDEEC (Enhanced Distributed Energy-Efficient Clustering). These protocols were selected for their effectiveness in cluster-based and energy-aware routing mechanisms.

The comparison was conducted using identical simulation parameters, ensuring a fair performance evaluation. Metrics assessed include Network Lifetime, Residual Energy, Packet Delivery Ratio (PDR), and Number of Dead Nodes across 500 rounds. EEPEG-PA-H consistently outperformed the benchmark protocols, especially in large-scale heterogeneous deployments.

EEPEG-PA-H achieved a longer network lifetime due to its priority-aware node selection and adaptive clustering.

The protocol maintained a higher average residual energy, reducing energy holes in the network. Packet delivery ratio was improved by efficient CH rotation and dynamic path optimization. Dead nodes appeared later in EEPEG-PA-H simulations, indicating more balanced energy consumption.

The results highlight that EEPEG-PA-H provides robust, energy-efficient, and scalable performance compared to these state-of-the-art protocols.

These results are graphically presented in Figure 4, such as Network Lifetime Comparison, Figure 5 as Residual Energy vs. Rounds, Figure 8 as Packet Delivery Ratio, and Figure 10 as Number of Dead Nodes.

4. CONCLUSION

This research introduces EEPEG-PA-H, a power-aware, adaptive routing protocol specifically designed to enhance the energy efficiency and extend the lifetime of Wireless Sensor Networks (WSNs) in the context of Internet of Things (IoT) applications. The protocol addresses one of the fundamental challenges in WSN-based IoT systems, energy conservation and network sustainability through intelligent resource management and routing optimization. The proposed strategy utilizes adaptive dynamic control, which includes the selective deactivation of redundant sensor nodes that report

overlapping, identical data. This approach significantly reduces energy waste while maintaining reliable data coverage. By integrating energy-efficient hierarchical clustering, probabilistic assessment, and power-aware multi-hop routing, EEPEG-PA-H effectively distributes energy loads across the network, thereby reducing early node failures and increasing overall network robustness. One of the key advantages of EEPEG-PA-H lies in its self-adaptive capability, which allows the protocol to respond dynamically to topological changes, node failures, and varying traffic patterns. This makes it highly suitable for large-scale and heterogeneous IoT deployments, where real-time sensing and long-term operation are essential. The protocol ensures consistent performance and prolonged network activity, even under fluctuating environmental and network conditions. Simulation results using NS demonstrate that EEPEG-PA-H outperforms traditional protocols such as PEGASIS and its modifications in terms of both network lifetime and energy consumption. The proposed protocol achieves up to 84.0% improvement in network lifespan, with energy reduction reaching up to 90.2%, validating its effectiveness and scalability in realistic IoT scenarios. In addition to performance gains, EEPEG-PA-H provides a flexible and modular framework that can be extended or integrated with emerging technologies. As part of future work, the protocol can be further enhanced by incorporating machine learning techniques for intelligent decision-making, predictive energy modeling, and anomaly detection. EEPEG-PA-H extends beyond prior work by integrating a unique priority-aware node selection mechanism, dynamic clustering, and adaptive routing strategies. The protocol is not a simple extension but a structured and multi-dimensional enhancement over PEGASIS and EEPEG-PA, offering significant improvements in energy efficiency, scalability, and reliability. Although the protocol is designed for potential use in dynamic environments, the present work does not include node mobility in the simulation. In the next phase, to integrate mobility models such as Random Waypoint or mobile sink mechanisms to evaluate the protocol's performance in dynamic and mobile WSN scenarios, including healthcare applications, and agree that adaptive thresholding is an important future direction. As part of ongoing research, to incorporate lightweight learning-based techniques to dynamically adjust thresholds in response to network context and application demands.

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Persian Abstract

چکیده

بهره وری انرژی یک عامل مهم در تقویت طول عمر عملیاتی شبکه های سنسور بی سیم (WSN) ، به ویژه در برنامه های اینترنتی اشیاء (IOT) است که در آن گره های حسگر محدود منابع باتری را محدود می کند .در این مقاله یک پروتکل پیشرفته ، جمع آوری کارآمد با انرژی با انرژی در سیستم های اطلاعاتی سنسور با خوشه بندی سلسله مراتبی قدرت (EEPEG-PA-H) ارائه می شود و یک ارزیابی مقایسه ای را در برابر پروتکل مسیریابی کسری ساده (SFRT) ارائه می دهد.پروتکل EEPEG-PA-H خوشه بندی پویا تطبیقی را با مسیریابی سلسله مراتبی آگاهی از برق یکپارچه می کند و باعث می شود تنظیمات در زمان واقعی در شکل گیری خوشه ای و مسیرهای مسیریابی بر اساس انرژی باقیمانده ، بار ترافیک و شرایط شبکه امکان پذیر شود .بر خلاف ماهیت استاتیک SFRT ، استراتژی پویا Eepeg-Pa-H به طور قابل توجهی انتقال های اضافی را کاهش می دهد و میزان مصرف انرژی را در گره ها متعادل می کند .نتایج شبیه سازی نشان می دهد که EEPEG-PA-H طول عمر شبکه را 23.33 ٪ نسبت به Pegasisو 1.34 ٪ نسبت به مدل EEPEG-PA قبلی بهبود می بخشد ، ضمن اینکه PDR 76 ٪ و توان 1.2 Mbps را حفظ می کند.این بهبود با کاهش قابل توجه تأخیر پایان به پایان به 190 میلی ثانیه و میزان از دست دادن بسته به 22 ٪ حاصل می شود .علاوه بر این ، عملکرد برتر در مقایسه با SFRT ، LEACH وVSSLS-SIACRاز نظر مصرف انرژی و پایداری شبکه را نشان می دهد .نتایج نشان می دهد که EEPEG-PA-H یک چارچوب مسیریابی قوی و کارآمد برای WSN های ناهمگن در برنامه های IoT ، با کاربرد بالقوه در محیط های پویا مانند مراقبت های بهداشتی موبایل فراهم می کند.